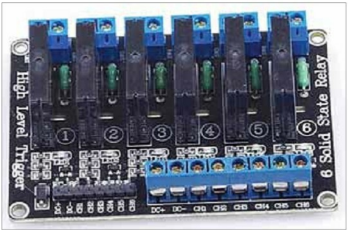

Fig. 3: Typical 5V, six-channel solid-state relay module

Fig. 4: CAN bus module

PARTS LIST	
Semiconductors:	
Board1-Board6	- Arduino Uno board
CM1-CM6	- MCP2515 CAN module
Resistors (all 1/4-watt, ±5% carbon):	
R1, R2	- 120-ohm
Miscellaneous:	
CON1	- 2-pin terminator connector
RM1-RM6	- 6-channel, 5V relay module
KP1-KP6	- 4×4 keypad
	- A shielded twisted pair cable
	- Jumper wires

need to include CAN_Bus_headers.zip and keypad.zip libraries in Arduino IDE.

Switching algorithm using Arduino Mega board

An Arduino Mega board has more digital pins to accommodate more switches, but the CAN bus has a limitation of five numeric digits. To tackle this problem, we reduce the node ID number to two digits (10, 11, 12, 13, ...99) instead of three digits (253, 254, 255, etc).

Therefore, while using Mega board, you can use a CAN bus ID, say 25. Then the switching signal with 25301# will switch on the relay connected to the digital pin D30 of Arduino Mega. Similarly, 25300# will switch off the relay connected to D30. You can thus have many more switches on Mega.

Construction and testing

Assemble the circuit shown in Fig. 2 on a breadboard. Connect all the six nodes via 2-pin cable: one for H cable and another for L cable. Make sure that the earth line (shielded) is also connected. Do not forget to connect 120-ohm resistor across H and L of CAN module. For short distance communication between two nodes, 120-ohm termination resistor is not required.

Note. One important point to prevent unknown operation on the node: Suppose one node is to be stopped from operation. But if it is unplugged from the bus, it will reset the bus and its node number will vanish from the system, which is not desirable.

So, the solution is to just disconnect the CS pin from the bus. It will remain on hold condition in that position. If some relays are already on, these will remain on. Similarly, if some relays are off, these will remain off. To restore, just reconnect the CS pin in the bus, which will start the normal operation again. EFY

EFY Note
The source code of this project is available for free download at *source.efymag.com*

Somnath Bera is an avid user of open source software. Professionally, he is a thermal power expert and works as additional general manager at NTPC Ltd